

Bank Account Management System using Classes and Objects

Objective / Aim

To write a Java program that demonstrates the concept of Classes and Objects by designing a Bank Account Management System. The program manages customer accounts and performs banking operations such as deposit, withdrawal, transfer, and interest calculation.

Algorithm / Logic Description

1. Start the program.
2. Create a class named Account with account details.
3. Create classes SavingsAccount and CurrentAccount.
4. Declare attributes such as account number, account holder name, balance, account type, interest rate, and overdraft limit.
5. Create methods for Deposit, Withdraw, Transfer, Calculate Interest, and Display Account Details.
6. Create objects for Savings Account and Current Account.
7. Perform deposit and withdrawal operations.
8. Calculate interest for the savings account.
9. Display updated account details.
10. Stop the program.

UML Class Diagram

Account
Account Number Account Holder Name Balance Account Type
Deposit() Withdraw() Transfer() Display details()

Current account	Savings Account
Interest Rate	Overdraft limit
Calculate interest()	Withdraw()

Source Code

```
class Account {  
    int accountNumber;  
    String accountHolder;  
    double balance;  
    Account(int accountNumber, String accountHolder, double balance) {  
        this.accountNumber = accountNumber;  
        this.accountHolder = accountHolder;  
        this.balance = balance;  
    }  
    void deposit(double amount) {  
        balance += amount;  
    }  
    void withdraw(double amount) {  
        balance -= amount;  
    }  
    void displayDetails() {  
        System.out.println("Account Number : " + accountNumber);  
        System.out.println("Account Holder : " + accountHolder);  
        System.out.println("Balance : " + balance);  
    }  
}  
  
class SavingsAccount extends Account {  
    double interestRate = 5;  
    SavingsAccount(int accountNumber, String accountHolder, double balance) {
```

```
        super(accountNumber, accountHolder, balance);
    }
    void calculateInterest() {
        double interest = (balance * interestRate) / 100;
        System.out.println("Interest = " + interest);
    }
}

class CurrentAccount extends Account {
    double overdraftLimit = 10000;

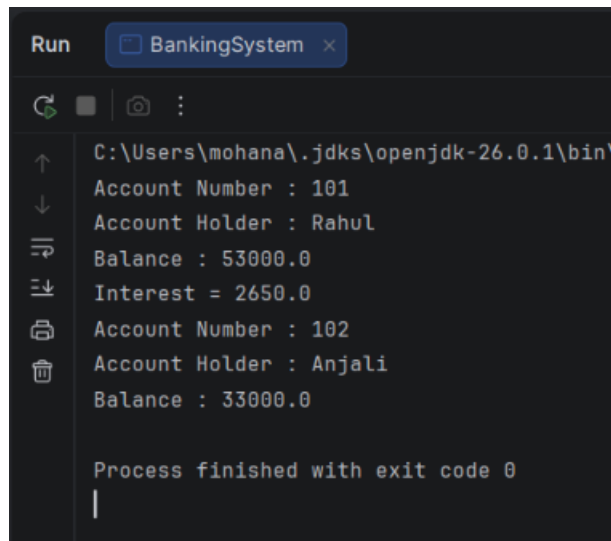
    CurrentAccount(int accountNumber, String accountHolder, double balance) {
        super(accountNumber, accountHolder, balance);
    }
}
```

```
public class BankingSystem {
    public static void main(String[] args) {
        SavingsAccount s = new SavingsAccount(101, "Rahul", 50000);
        s.deposit(5000);
        s.withdraw(2000);
        s.displayDetails();
        s.calculateInterest();

        CurrentAccount c = new CurrentAccount(102, "Anjali", 30000);
        c.deposit(4000);
        c.withdraw(1000);
        c.displayDetails();
    }
}
```

}

Output Screen :



```
Run  BankingSystem x
C:\Users\mohana\.jdk\openjdk-26.0.1\bin\
Account Number : 101
Account Holder : Rahul
Balance : 53000.0
Interest = 2650.0
Account Number : 102
Account Holder : Anjali
Balance : 33000.0

Process finished with exit code 0
|
```